

Entity Relationship Concepts

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1. Formal Definition

An Entity is any real-world object, person, place, event, concept, or thing about which the database stores information.

2. Characteristics of an Entity

Every entity has certain properties:

1. It has an identity: Each entity can be uniquely identified.

2. It has attributes: Every entity has properties.

Example (Student): Name, Age, Roll Number.

3. It exists independently in the problem domain: A student exists whether or not they are enrolled in a course.

4. Multiple instances may exist:

Entity: Student

Instances: Arpit, Aadya.

3. Entity Set

An Entity Set is a collection of similar entities of the same type having same attributes so arpit+aadya = student entity set.

4. Entity Types

Strong Entity

A Strong Entity can exist independently and has its own primary key.

Weak Entity

A Weak Entity is an entity that cannot be uniquely identified by its own attributes alone and depends on another (owner) entity for its identity and existence.